

Corso: Large Language Models for Software Engineering

Quiz: Exam (2026-06-19)

Iniziato	19 giugno 2026, 09:10
Stato	Completato
Terminato	19 giugno 2026, 10:24
Tempo impiegato	1 ora 14 min.
Valutazione	12,13 su un massimo di 18,00 (67%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11

Domanda 1

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

You are given the following sequence of characters.

c c b a a a d a a c c c

Initially, each character is a separate token.

Apply Byte-Pair Encoding to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots.

Separate the merged tokens with a comma (and no spaces).

Examples:

- 1) if the merged tokens are aa and b, write: aa,b
- 2) if the merged tokens are x and y, write: x,y

1st merge:

Risposta corretta
La risposta corretta è : a,a
Punteggio ottenuto 1,00 su 1,00

2nd merge:

Risposta corretta
La risposta corretta è : c,c
Punteggio ottenuto 1,00 su 1,00

3rd merge:

Risposta corretta
La risposta corretta è : aa,a
Punteggio ottenuto 1,00 su 1,00

Domanda 2

Completo

Punteggio
ottenuto 3,00
su 3,00

(3 points, no penalties for wrong answers)

Consider a scenario where a software team wants to use LLAMA-based agents to help improve a pull request.

You are given:

- FEATURE_SPEC: a natural language description of the feature
- PR_CODE: the code submitted in the pull request
- TEST_RESULTS: the results of running tests on the code, including passed and failed tests

Design a simple agentic workflow that uses LLAMA-based agents to:

- Understand the feature requirements
- Improve the code in the pull request

- Use test feedback to refine the code if tests fail

For each agent, briefly specify:

- Its role
- Its system prompt
- Its input
- Its output

You do not need to write code or actual LLAMA calls. Focus only on the agents, their prompts, and how information flows between them.

Architecture:

For the architecture i am using chain of thoughts which consist in concatenate three agents , where the second and the third depend on the previous output. Also i will use RGC technique (Role, Goal, Context) with zero-shot approach.

prompt1=""

ROLE=You are an expert in pull request and in software engineering. You have to give me a hand with some documents that i will send you in the input.

GOAL=Your goal is to understand the feature requirements, Improve the code in the pull request and Use test feedback to refine the code if tests fail.

CONTEXT=You can imagine to work in a sotware team using GitHub or VSCode as a environment programming.

input = FEATURE_SPEC which is a natural language description of the feature, PR_CODE which is the code submitted in the pull request and TEST_RESULTS which are the results of running tests on the code, including passed and failed tests.

output1 = Give me as an output the input files with the context

prompt2=""

ROLE= You are an expert in reviewing features, pull request and software engineering.

GOAL = Your goal is to take in input the output of the prevoius agent <output1> and review if all the work which is understand the feature requirements, Improve the code in the pull request and Use test feedback to refine the code if tests fail , is all DONE.

CONTEXT = You can imagine to work in a sotware team using GitHub or VSCode as a environment programming.

input = take the output<output1> of the first agent and control if all the work is well done.

output2=Give me as an output the input files with the context.

prompt3=""

ROLE= You are an expert in reviewing features, pull request and software engineering.

GOAL = Your goal is to take in input the output of the prevoius agent <output2> and review if all the work which is understand the feature requirements, Improve the code in the pull request and Use test feedback to refine the code if tests fail , is all DONE.

CONTEXT = You can imagine to work in a sotware team using GitHub or VSCode as a environment programming.

input = take the output<output2> of the first agent and control if all the work is well done.

output3=Give me as an output the input files with the context.

Commento:

Domanda 3

Risposta
corretta

1 point (-25% penalty for each wrong answer)

What is "priming" in the context of prompt engineering?

Punteggio
ottenuto 1,00
su 1,00

- ☒ a. Providing initial input to guide the model's response ✓
- ☐ b. Resetting the model before every new prompt
- ☐ c. Using reinforcement learning to fine-tune the model
- ☐ d. Pre-training the model with additional data
- ☐ e. Adding noise to the input to test model robustness

Risposta corretta.

La risposta corretta è:

Providing initial input to guide the model's response

Domanda 4

Risposta
errata

Punteggio
ottenuto -0,38
su 1,50

1.5 points (-25% penalty for a wrong answer)

The following reward can be used when fine-tuning a model on human feedback.

$$R(x, y) = r_{\theta}(x, y) - \beta \log \frac{\pi_{new}(y|x)}{\pi_{old}(y|x)}$$

Where r_{θ} is the output of a Reward Model, π_{new} and π_{old} are the new and the original policies, respectively (you can see them as the probabilities assigned by the new and original models).

What is the purpose of the term $\log \frac{\pi_{new}(y|x)}{\pi_{old}(y|x)}$?

- ☐ a. it helps make the model more deterministic
- ☐ b. it improves the model's capability of following instructions
- ☐ c. it rewards the model when it generates an answer that humans will "like"
- ☐ d. none of the other answers is correct
- ☐ e. it prevents the model from collapsing into a model that always predicts a single token
- ☐ f. it increases the entropy of the output (i.e., it helps the model produce more "creative" answers)
- ☒ g. it prevents the new model from producing outputs that are too similar to the original one ✗

Risposta errata.

La risposta corretta è: none of the other answers is correct

Domanda 5

Risposta
errata

Punteggio
ottenuto 0,00
su 1,50

1.5 points (-15% penalty for a wrong answer)

You are given an **encoder-only** model. What is the total number of parameters (token + positional + transformer layers + language model head)?

Note: assume that none of the layers uses a bias term. You can ignore layer normalization parameters.

- Vocabulary size $V=29,000$
- Embedding/model dimension $d=256$
- Positional embeddings: absolute, learned, for 512 positions
- Number of layers $L=9$
- Number of attention heads $H=16$

- FF-NN projection dimension: 32
- Key, query size $d_k = 4$
- Value size $d_v = 64$

Choose the correct answer. Round the result to the closest million.

Risposta: ❌

La risposta corretta è : 20

Domanda 6

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (-25% penalty for a wrong answer)

In a Transformer's self-attention mechanism, what is the primary purpose of the matrices W_K , W_Q , and W_V ?

- ☒ a. They are used to transform the input token embeddings into vectors that are passed to the attention mechanism ✓
- ☐ b. They are used to store the query, key, and value vectors learned during training for each token
- ☐ c. They are passed to the attention mechanism to compute attention scores between tokens
- ☐ d. None of the other answers is correct
- ☐ e. They are used to transform positional encodings into query, key, and value vectors before attention
- ☐ f. They are used to transform the output of the attention mechanism back into token embeddings
- ☐ g. They are used to combine the outputs of multiple attention heads into a single representation

Risposta corretta.

La risposta corretta è:

They are used to transform the input token embeddings into vectors that are passed to the attention mechanism

Domanda 7

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

You want to fine-tune a decoder-only transformer-based model using LoRA. The details of the model are reported below.

- Vocabulary size $V=74,000$
- Embedding/model dimension $d=4096$
- Positional embeddings: absolute, learned, for 1024 positions
- Number of layers $L=20$
- Number of attention heads $H=8$
- FF-NN projection dimension: 1024
- Key, query size $d_k = 128$
- Value size $d_v = 256$

You apply LoRA to the W_K , W_Q and W_V matrices. You want to fine-tune **at most 20 M** parameters. What is the highest possible low-rank dimensionality to be used?

Assume that each of the attention heads has separate matrices for W_K , W_Q and W_V (many implementations stack the matrices for all heads into a single matrix -- for this exercise, assume that each

matrix is separate from the others).

Do not consider bias terms in the computations. Write the answer in the box below.

Risposta: ✓

La risposta corretta è : 9

Domanda 8

Completo

Punteggio
ottenuto 2,00
su 3,00

(3 points - no penalties for wrong answers)

You are evaluating an LLM that extracts functional requirements from a specification for an online food delivery platform. The LLM is given a paragraph of system requirements and produces a list of extracted candidate requirements.

Specification:

"Customers can search restaurants by cuisine and location, view restaurant menus, add dishes to a cart, and place orders for delivery. The system allows users to track the order status in real time and pay online using credit cards or digital wallets. Restaurants must be able to accept or reject incoming orders. Delivery drivers should receive notifications when a new delivery is assigned to them."

From this, the ground truth requirements are:

- G1: Search restaurants by cuisine and location
- G2: View restaurant menus
- G3: Add dishes to a cart
- G4: Place delivery orders
- G5: Track order status in real time
- G6: Pay online using credit cards or digital wallets
- G7: Allow restaurants to accept or reject incoming orders
- G8: Notify delivery drivers when a delivery is assigned

Requirements extracted by the LLM:

- L1: Find restaurants based on cuisine
- L2: Browse available menus
- L3: Add food items to a shopping cart
- L4: Reserve a table at a restaurant
- L5: Place an order for home delivery
- L6: Track delivery progress live
- L7: Pay using PayPal only
- L8: Send alerts to drivers for assigned deliveries
- L9: Allow restaurants to cancel customer accounts

Identify the following:

- True Positives (TP)
- False Positives (FP)
- False Negatives (FN)

Compute:

- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
- F1-Score = $2 \times (P \times R) / (P + R)$

You can write in free text your assumptions about the matching between ground truth requirements and LLM-extracted requirements.

$$TP = 6$$

True Positive are extracted features that matches ground truth

$$FP = 3$$

False Positive are extracted features but doesn't matches ground truth

$$FN = 0$$

False Negative are wrong extracted features of the ground truth

$$\text{Precision} = 6 / (6+3) = 0.6667$$

$$\text{Recall} = 6 / (6 + 0) = 1$$

$$F1 - \text{Score} = 2 \times (0.6667 \times 1) / (0.6667 + 1) = 0.8000$$

Commento:

there are two false negatives

Domanda 9

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

Which of the following are benefits of a Retrieval-Augmented-Generation approach to prompting?

- ☒ a. It is possible to reduce the amount of false information provided to the final user. ✓
- ☐ b. It is possible to adopt self-critique mechanisms to refine the outputs of the agent.
- ☒ c. It is possible to prevent the obsolescence of answers by using up-to-date documents. ✓
- ☐ d. It is possible to reduce the amount of hallucinations by providing a fixed structure for the agent's answers.
- ☐ e. It is possible to decompose a task in several smaller steps.

Risposta corretta.

Le risposte corrette sono:

It is possible to reduce the amount of false information provided to the final user.,

It is possible to prevent the obsolescence of answers by using up-to-date documents.

Domanda 10

Risposta
errata

Punteggio
ottenuto 0,00
su 1,50

1.5 points (no penalty for a wrong answer)

The BLEU-n score is computed as follows:

$$\text{BLEU-n} = BP \cdot \exp\left(\frac{1}{n} \sum_{i=1}^n \log p_i\right)$$

Where p_i is the precision for i-grams (i.e., the fraction of matching i-grams between reference and generated sequences, w.r.t. the number of i-grams in the generated sequence).

BP is the brevity penalty, defined as:

$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}$$

Where c is the length of the generated sequence and r is the length of the reference sequence.

You are given the following sequences, already divided into tokens:

Reference <tok0> <tok8> <tok11> <tok5> <tok9> <tok4> <tok1> <tok2> <tok7> <tok19>

Generated <tok3> <tok17> <tok1> <tok2> <tok7> <tok19> <tok5>

Compute the corresponding BLEU-2 score. Write the answer in the box below, reporting at least 4 digits after the decimal point.

Risposta: ✖

La risposta corretta è : 0,38930929939948

Domanda 11

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for wrong answers)

In agentic behavior, what changes compared to a traditional fixed workflow?

- ☐ a. The model must always use RAG
- ☒ b. The model decides its own control flow and when to stop ✔
- ☐ c. The model follows a predetermined control flow
- ☐ d. The model can't use tools
- ☐ e. The model always does multiple-step tasks

Risposta corretta.

La risposta corretta è:

The model decides its own control flow and when to stop